

DEGREE: Computer Science and Digitisation

Module: Algorithms and Problem Solving using Python

Assignment Title: Transforming a Real world problem into algorithm using Python Program

Assignment Type: Individual report

Word Limit: 2000 words (+/- 200)

Weighting: 50%

Issue Date: 28/04/2026

Submission Date: 15/06/2026

Feedback Date: 29/06/2026

Issued by: Dr. Syed Arslan Abbas Rizvi

Plagiarism:

When submitting work for assessment, students should be aware of the InterActive/Canvas guidance and regulations in concerning plagiarism. All submissions should be your own, original work. Please note that you must not submit the same assignment for two different modules within your course.

You must submit an electronic copy of your work. Your submission will be electronically checked.

Learner declaration		
Word count		
Use of proof-reader/proof-reading service (see guidance contained in Academic Misconduct Regulations on appropriate use)	YES	NO

If 'yes', please specify which service (e.g. Grammarly, Studiosity etc.)		
I confirm that I submit this work as my own work and that I have cited all sources I have used, and I understand that using sources without citing them correctly may be considered as Academic Misconduct.	YES	NO
I confirm that I have followed guidance on the acceptable use of AI tools for this assignment where such guidance has been issued by my tutors.	YES	NO
Where use of appropriately cited use of AI tools is permitted, I confirm that I have cited in accordance with the UCA Harvard Referencing Standard.	YES	NO

Harvard Referencing:

The Harvard Referencing System must be used. The Wikipedia, UKEssays.com or similar websites must **not** be used or referenced in your work.

By submitting this assessment, you are confirming that you have understood and are complying with the University's Academic Misconduct Regulations.

Plagiarism Detection (Turnitin)

BSBI uses Turnitin (an on-line text matching tool) to help identify cases of plagiarism. Turnitin will compare a submitted assignment with a database of journal articles, books, web pages and previously submitted assignments.

Signature (Student): _____ Date: _____

Introduction

Learning Outcomes:

LO1. Demonstrate an understanding and examine how different data structures and algorithm design methods including lists, stacks, queues, trees, and graphs impact the performance of programs

LO2. Implement the algorithms for problem solving using Python code

LO3. Implement and execute Python programs using functions, modules, libraries and classes

Assessment Criteria: Weighting 50%

2000 words

Objective:

The objective of this assignment is to demonstrate proficiency in algorithms, data structures, and problem-solving techniques using Python. Through practical implementations, you will explore various data structures, algorithm design methods, and Python programming features to solve real-world problems effectively.

Theoretical and Practical Implementation and Analysis of Sorting Algorithms

You are working on a real-world data processing task from a library management system. The system logs borrowing records in a plain text file, where each record contains:

A borrower's name (string) and the number of days the item was kept (integer). An example of this dataset is shown below.

Borrower's Name	Emma	Liam	Sophia	Mason	Olivia	Noah	Ava
Days since Issued	23	5	31	14	7	42	18

The library requires the records to be sorted by the number of days (ascending) to generate an "overdue report." However, the system must handle varying input sizes from small daily logs (~10–50 records) to monthly exports (~10,000+ records). You are required to select an optimum sorting algorithm suited for this task and design a programme in python to solve this problem for the library.

Deliverables:

Implement a data sorting solution in Python by reading a file containing the library dataset. For demonstration you can create a made-up test dataset with the format given above. Consider all the following:

- Analyse time and space complexity (Big O notation) for different sorting methods considering the best, average, and worst cases. Show with examples for small dataset and large dataset to justify your selection of sorting algorithm.
- Identify what data structures and data types are suitable for handling the type of data described in the above scenario.
- **Draw a flowchart** with proper description of each step clearly explained in your own words should be included in your answers.
- A **python code** with proper comments should be submitted along with the assignment.
 - The program should identify who are the people who are late in returning the book in its output.
 - the code should also show number of records sorted, running time, number of computations (logical, arithmetic, swapping etc) required for sorting and identifying the late returns.
- Conclude your report with all your findings about sorting algorithm choice, its performance as compared to other algorithms, its limitations and advantages.
- Future improvements that can be implemented in this code.

Submission Guidelines:

- Submit a well-documented Python script (in the form of a listing or appendix) containing your implementations, analysis, and explanations in you assignment.
- Use good visualizations (figures and tables) with proper captioning and cross referencing in your report. No figures / tables should be left unexplained.
- Ensure that your code is properly commented to explain the logic and algorithms used.
- Include any additional documentation or insights gained during the implementation process.
- Prepare a document using the BSBI assignment template available in Canvas.
- Use Harvard referencing style for your bibliography.
- Refer to the Essay-Guide available in Canvas for further instructions.
- Submit your assignment electronically by the specified deadline.

Students are encouraged to explore additional reputable sources in the digital library beyond the list provided to strengthen their individual research and critical analysis skills. The unit lecturer can help with recommending additional literature.

The criteria for successfully passing the units with two assessment components require passing both components successfully. Passing marks for BSc and MA/MSc levels are different and are stated at the end of the assignment brief. Please find the examples below:

- **Example 1** (e.g., pass mark: 50)
 - Component 1 (50% weighting) - 55
 - Component 2 (50% weighting) - 55
 - Final average: 55
 - Result: **Pass (both components passed and therefore the unit is passed)**

- **Example 2** (e.g., pass mark: 50)
 - Component 1 (50% weighting) – 80
 - Component 2 (50% weighting) – 30
 - Final average: 55
 - Result: **Fail (because although the average is above 50, one of the components is a failure)**

GUIDANCE ON ASSESSMENT

All materials must be properly referenced under Harvard conventions. The length required is **2000** words with tasks equally weighted. The writing style should be formal academic / report writing style with in-text referencing to support your comments and observations. Originality, quality of argument and good structure are required. The student should demonstrate sound understanding and ability to apply knowledge and theory of **Algorithms and Problem Solving using Python**. Additional marks being awarded for juxtaposition and insight of issues.

Grading Criteria

	Generic Criteria	90 - 100	80 - 89	70 - 79	60 - 69	50 - 59	40 - 49	30 - 39	0 - 29
Level 4	Knowledge of contexts, concepts, technologies and processes The extent to which knowledge is demonstrated: <i>relevant contextual or theoretical issues are identified, defined and described</i> <i>historical or contemporary practices are identified, defined and described</i> <i>appropriate technologies, methods and processes are identified, defined and described</i>	Outstanding breadth of knowledge of fundamental contextual and theoretical issues and critical concepts and their relationship to historical and contemporary practices Extensive knowledge of relevant and specialist technologies and processes	Extensive knowledge of fundamental contextual and theoretical issues and critical concepts and a widening appreciation of historical and contemporary practices In depth and broadening knowledge of appropriate technologies and processes	Significant knowledge of fundamental contextual and theoretical issues and critical concepts and a widening appreciation of historical and contemporary practices Significant knowledge of appropriate technologies and processes	Confident familiarity with fundamental contextual and theoretical issues and critical concepts Thorough knowledge of appropriate technologies and processes	Familiar with fundamental contextual and theoretical issues and critical concepts Sound knowledge of appropriate technologies and processes	Adequate knowledge of fundamental contextual and theoretical issues and critical concepts Adequate knowledge of appropriate technologies and processes	Limited knowledge of fundamental contextual and theoretical issues and critical concepts Limited knowledge of appropriate technologies and processes	Little or no knowledge of fundamental contextual and theoretical issues or critical concepts Little or no knowledge of appropriate technologies or processes
	Understanding through application of knowledge The degree to which research methods are demonstrated: <i>relevant knowledge and information is compared, contrasted, manipulated, translated and interpreted</i> <i>knowledge and information is selected, analysed, synthesised and evaluated in order to generate creative ideas, practices, solutions, arguments or hypotheses</i>	Relevant knowledge is explored and interpreted when proposing solutions to projects and problems which demonstrate evidence of independent thought Outstanding ability to analyse and synthesise knowledge to produce own creative practice in standard situations and to evaluate results	Deep level of comprehension and exploration of relevant knowledge in seeking solutions to projects or problems Outstanding ability to analyse and synthesise knowledge in order to produce creative practice in standard situations and to evaluate the results	Deep level of comprehension of relevant knowledge in seeking solutions to projects or problems Strong ability to apply and analyse knowledge to produce creative practice in standard situations, with some evaluation of the results	Strong comprehension of relevant knowledge in seeking solutions to projects or problems Sound ability to apply and analyse knowledge to produce creative practice in standard situations	Sound comprehension of relevant knowledge in seeking solutions to projects or problems Sound ability to apply knowledge to produce creative practice in standard situations	Surface-level comprehension of relevant knowledge in seeking solutions to projects or problems Competent application of knowledge to the production of creative practice in standard situations	Incomplete comprehension of relevant knowledge in seeking solutions to projects or problems Limited ability to apply knowledge to produce creative practice in standard situations	Little or no comprehension of relevant knowledge in seeking solutions to projects or problems Little or no ability to apply relevant knowledge to produce creative practice in standard situations
	Application of technical and professional skills The degree to which:	Accomplished and fluent application of appropriate practical and technical skills	Relevant, accomplished and fluent application of basic practical and technical skills	Relevant and accomplished application of basic practical and technical skills	Strong application of basic practical and technical skills	Sound application of basic practical and technical skills	Competent application of practical and technical skills	Rudimentary application of basic practical and technical skills	Scant application of basic practical and technical skills

<i>appropriate materials and media are selected, tested and utilised to realise and present ideas and solutions</i> <i>appropriate technologies, methods and processes are demonstrated</i> <i>transferable, professional skills are effectively demonstrated</i> <i>self management and independent learning are demonstrated</i>	Outstanding application of appropriate transferable and professional skills Significant ability to learn independently and critically evaluate own progress using a wide range of feedback sources	Outstanding application of fundamental transferable and professional skills Substantial ability to work independently and use feedback to reflect critically on own progress	Highly effective application of fundamental transferable and professional skills Strong ability to work independently and use feedback to plan future tasks effectively	Strong application of fundamental transferable and professional skills Evidence of developing well as an independent learner	Sound application of fundamental transferable and professional skills Evidence of beginning to develop as an independent learner	Competent application of fundamental transferable and professional skills Adequate evidence of beginning to develop as an independent learner	Limited application of fundamental transferable and professional skills Limited evidence of ability to learn independently	Ineffective application of fundamental transferable and professional skills Little or no evidence of ability to learn independently
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